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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO5 ASSESSMENT TOOL 3

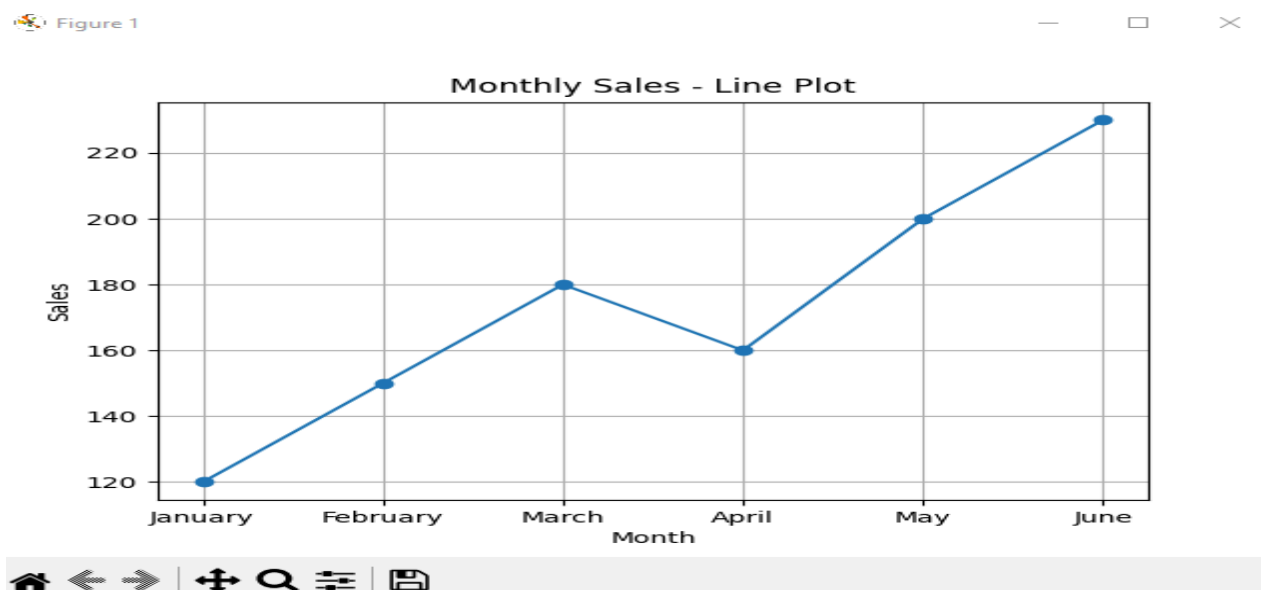
20.08.2026

1. COMPARE PLOTS TYPES : Use the same dataset to create a line plot, bar plot and scatter plot. Discuss which plot communicates the data most effectively.

AIM: To compare line, bar, and scatter plots using the same dataset and determine which plot communicates the data most effectively.

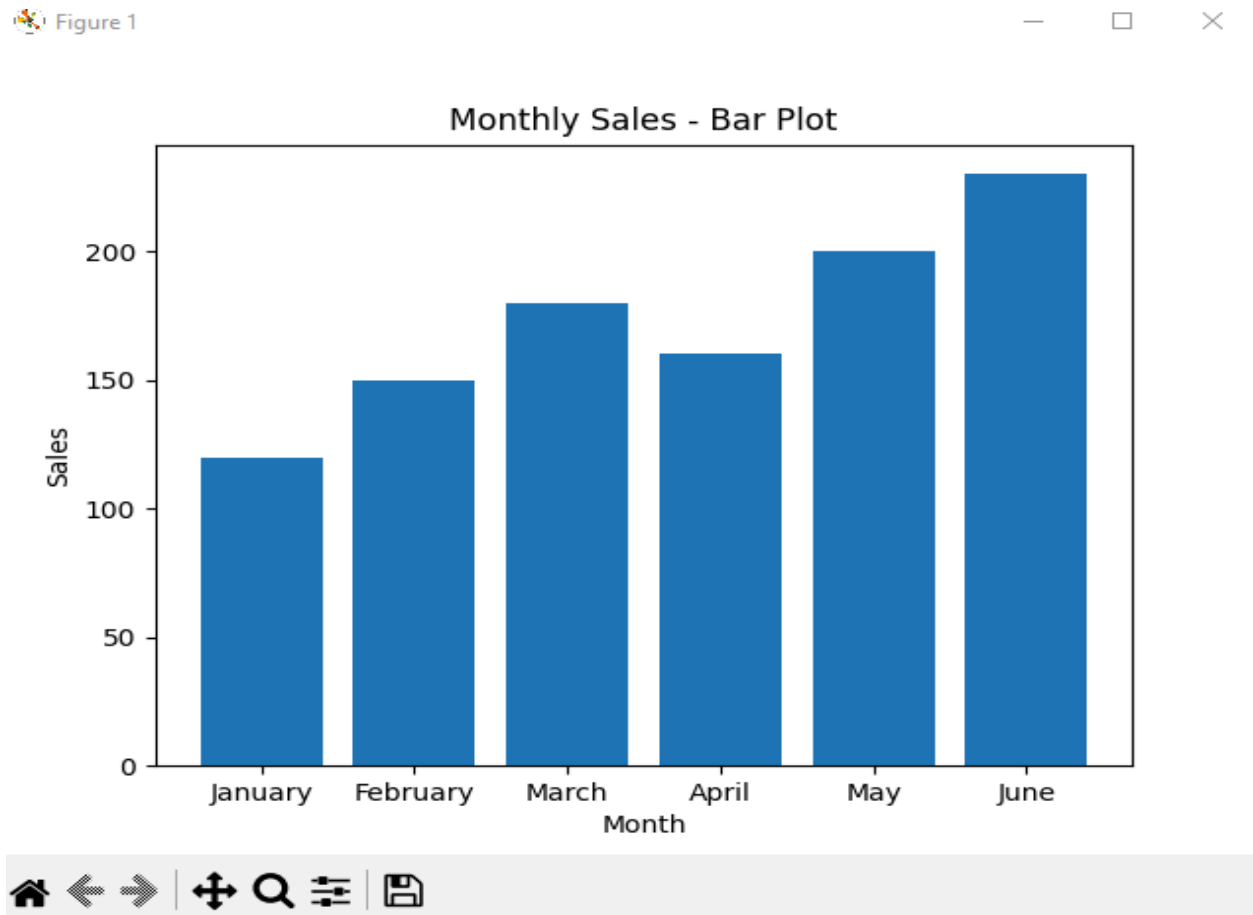
DATASETS : MONTHLY SALES DATASETS

LINE PLOT :



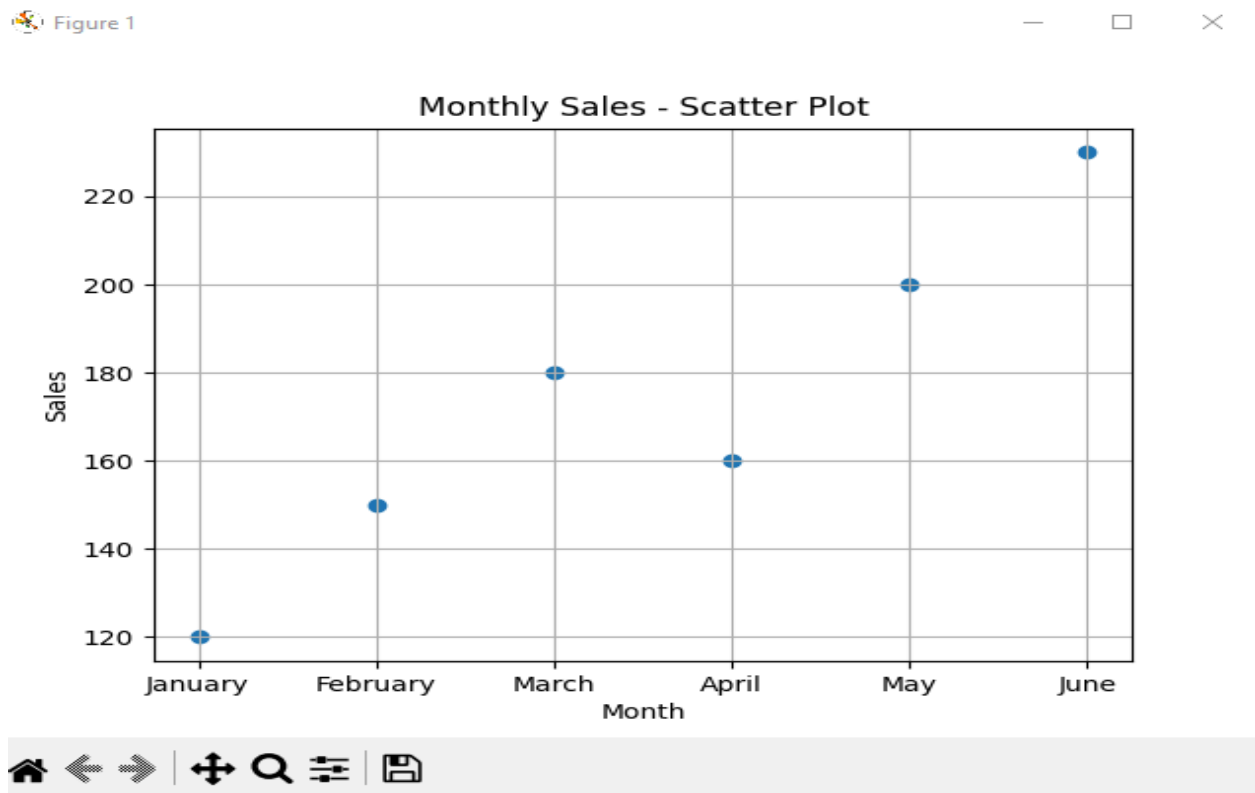
The line plot shows the change in sales over the months. It clearly represents the increasing and decreasing trends.

BAR PLOT :



The bar plot makes it easy to compare the sales values of each month.

SCATTER PLOT :



The scatter plot displays individual sales observations and helps visualize the relationship between the month and sales.

JUSTIFICATION :

The line plot communicates the data most effectively because the dataset represents monthly sales over time. It clearly shows the overall trend and changes in sales. The bar plot is useful for comparing individual months, while the scatter plot is useful for observing relationships between variables.

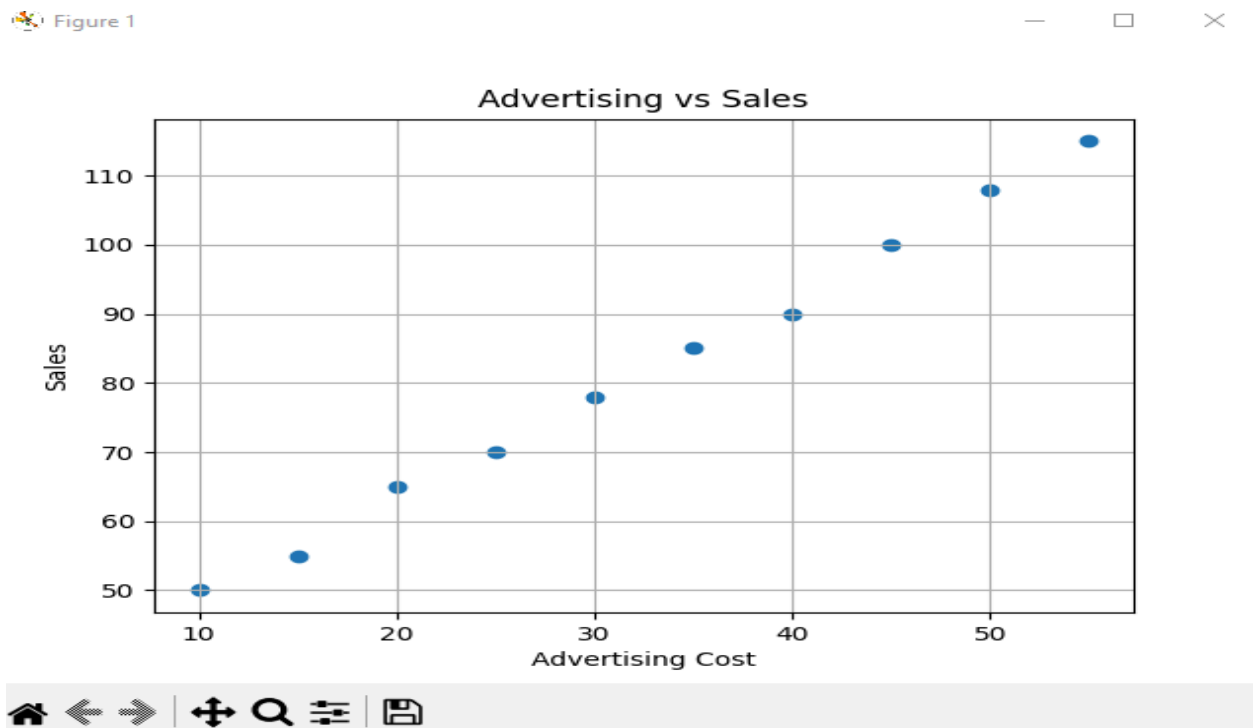
CONCLUSION : The three plot types represent the same sales dataset in different ways. The line plot is the most suitable for showing trends over time, while the bar plot is best for comparing values and the scatter plot is useful for examining relationships.

2. Relationship Investigation : Plot two variables such as Advertising vs Sales using a scatter plot. Determine whether the relationship appears positive, negative or weak

AIM: To investigate the relationship between advertising expenditure and sales using a scatter plot and determine whether the relationship is positive, negative, or weak.

DATASET : ADVERTISING VS SALES

OUTPUT:



Advertising increases → Sales also increase

The points will generally move upward from left to right.

Therefore: Relationship = Positive

JUSTIFICATION : The scatter plot shows a positive relationship between advertising expenditure and sales. As advertising expenditure increases, sales also tend to increase. The points generally move upward from left to right, indicating a positive relationship.

CONCLUSION: The investigation shows that advertising and sales have a positive relationship. Higher advertising expenditure is associated with higher sales in the given dataset.